

Exam. Level Programme Year / Part	Regular (2066 & Later Batch)		
	BE	Full Marks	80
	BCT	Pass Marks	32
	III / II	Time	3 hrs.

Subject: - Artificial Intelligence (CT 653)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Assume suitable data if necessary.

1. Discuss any two fields of your daily life where artificial intelligence has been applied. (7)
2. Solve the following crypto-arithmetic problem, where different letters denote different integers and identical letters denote same integer. $LOGIC + LOGIC = PROLOG$
Show all the step of solving through constraint satisfaction problem. (7)
3. Discuss the hill-climbing search algorithm along with problems associated with it and discuss their solutions. (9)
4. Given premises: Every American who sells weapons to hostile nations is a criminal. The country Abc is enemy of America. All of the missiles in Abc were sold by John. John is an American.
Proof: John is a criminal. (10)
5. What are the different knowledge representation models? Discuss semantic nets with an example. (7)
6. What is Fuzzy learning? Explain with a practical example. (4)
7. Explain the learning framework with suitable example. (6)
8. What is a Hopfield Network? Explain all the steps involved in the Hopfield Network with suitable example. (8)
9. Explain different steps of expert system development with an example. (8)
10. What is a natural language processing? Explain it. (6)
11. Write short notes: (any two) (4 x 2 = 8)
 - i. Skolemization
 - ii. Machine vision
 - iii. Human Brain verses Neural Network

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- 4] 1. What is artificial intelligence (AI)? Explain the Turing and its importance in artificial intelligence. [7]
- 2] 2. Solve the following crypto-arithmetic problem, where different letters denote different integers and identical letters denote same integer. [7]
2] RIGHT + RIGHT = WRONG
Show all the step of solving through constrain satisfaction problem.
3. What is blind search? Explain depth first search algorithm with an example and compare it with breadth first search. [9]
4. Explain all the steps involved in conversion of conjunctive normal form (CNF) with an example. [10]
5. What is a frame? Explain it with suitable example. [7]
6. What is genetic algorithm (GA)? Explain operator of GA with its importance. [10]
7. What is neural network? Differentiate between supervised and unsupervised learning. [6]
8. What is expert system? Explain general architecture of an expert system. [8]
9. What is natural language processing (NLP)? Explain different steps of NLP. [8]
10. Write short notes on: (any two) [4×2]
 - a) Horn clause
 - b) Machine vision
 - c) Perceptron

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1. Define AI. When a machine is said to be passed Turing test? Give any two examples of constraint satisfaction problem. [2+5+1]
2. Solve the following crypto-arithmetic problem, where different letters denote different integers and identical letters denote same integer. WRONG + WRONG = RIGHT. Explain the steps that you have followed. [5+3]
3. Differentiate between informed and blind search. How depth search is different to breadth first search. Compare with evaluation parameters. [4+4]
4. All oversmart persons are stupid. Children of oversmart persons are naughty. Ram is children of Hari. Hari is oversmart. Show that Ram is naughty. Using FOPL based resolution method. [8]
5. Explain the step involved in conjunctive normal form (CNF) with suitable example. [8]
6. What is semantic net? Explain with suitable example. [8]
7. What is machine vision? Discuss about the algorithm of Genetic Algorithm. [2+6]
8. What is neural network? Explain back-propagation algorithm learning.. [4+4]
9. What is an Expert System? Explain the steps of an Expert System development. [4+4]
10. Define machine translation in NLP. Explain the challenges of machine translation. [1+7]

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1. What is Artificial Intelligence (AI)? Discuss brief history of AI with Chronological development. [2+6]
2. Why searching is necessary in AI? Explain about the role of production system with suitable example. [2+6]
3. What is horn clause? Differentiate between Depth First Search and Breadth First Search. [1+7]
4. Explain backward chaining with suitable example and compare with forward chaining. [4+4]
5. Why do we need FOPL? State any three rules of inference. How can we make the machine with learning capacity? [2+3+3]
6. Define Boltzmann Machine. How knowledge can be represented using semantic network? Explain with suitable example. [1+7]
7. What is Machine Learning? What is Fuzzy Logic? Explain the Fuzzy Inference with suitable example. [2+6]
8. Differentiate declarative knowledge and procedural knowledge. Explain the architecture of expert system. [2+6]
9. What is the role of perceptron in neural network? Explain about bakpropagation algorithm. [3+5]
10. What is Natural Language Processing (NLP)? Discuss the different issues related with NLP with example. [2+6]

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1. What is Artificial Intelligence (AI)? Discuss history of AI in brief. [2+5]
2. What is the advantage of depth limit search? Compare it with other search strategies. [4+4]
3. If X is on the top of Y, Y supports X. If X is above Y and they are touching each other, X is on the top of Y. A cup is above a book. A cup is touching a book. Show that supports (book, cup) is true. [8]
4. Give an example of learning by analogy. How knowledge can be represented using semantic network? Explain with suitable example. [1+7]
5. Explain Backward Chaining with suitable example. [7]
6. What is conceptual dependency? Explain some of common primitives used in conceptual dependency. [2+5]
7. What makes a problem "Well Defined"? Explain with a sample example of a state-space search framework. [3+4]
8. What is Artificial Neural Network (ANN)? Compare ANN with human brain with its functioning principle. [3+5]
9. Define knowledge acquisition with example. Explain the architecture of expert system. [2+6]
10. Compare the following: [4×3]
 - a) Declarative versus Procedural Knowledge
 - b) Pragmatic versus Phonetic Analysis
 - c) Genetic algorithm based versus Fuzzy based learning

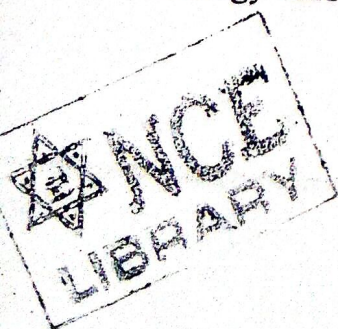
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1. Distinguish between knowledge and learning. What does acting humanly refer to? Explain. Define well defined problems. [2+5+1]
2. Solve the following cryptoarithmic problem with necessary steps. [7]

$$\text{ONE} + \text{ONE} + \text{TWO} = \text{FOUR}$$
3. How informed search is different from uniformed search? Explain min-max algorithm with suitable example, also discuss how alpha-beta is different from min-max algorithm. [2+4+3]
4. What is causal network? Explain reasoning in belief network with suitable example. [2+5]
5. What is neural network? Describe its types. A doctor knows that the disease meningitis causes the patient to have a stiff neck 50% of the time. The doctor also knows that the probability that a patient has meningitis is $1/50,000$, and the probability that any patient has a stiff neck is $1/20$. Find the probability that a patient with a stiff neck has meningitis? [3+5]
6. What is machine learning? Explain genetic algorithm along with different operators of genetic algorithm. [2+8]
7. What is Natural Language Processing (NLP)? Explain the syntactic, semantic and pragmatic analysis of NLP. [2+2+2+2]
8. Define expert system? Discuss about the general architecture of an expert system. [2+4]
9. How does machine vision help in Artificial Intelligence? Explain how back propagation algorithm helps in machine learning. [3+5]
10. Compare the followings: [3x3]
 - a) Forward versus backward chaining
 - b) Hopfield versus Kohonen network
 - c) Analogy versus Inductive learning



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1. What is an Artificial Intelligence (AI)? Explain any two applications of AI in real field. [7]
2. What do you understand by Constraint satisfaction problem? Solve the following Crypt-arithmetic problem. [1+6]

SEND
+ MORE

MONEY

3. What is a searching? Explain Breadth First Search and Depth First Search and compare their performance criteria. [9]
4. What is a knowledge, representation and reasoning? Describe forward chaining with practical example. [2+5]
5. Assume the following facts: [7]
 - John likes all kinds of food.
 - Apples are food.
 - Chicken is food.
 - Anything anyone eats and isn't killed by is food.
 - Bill eats peanuts and is still alive.
 - Sue eats everything Bill eats.

Prove that John likes peanuts using resolution refutation

6. What are semantic nets and frames? How frames are useful in semantic nets. [7]
7. What is a machine learning? Explain in detail about Boltzmann machines with suitable algorithm and explanations. [2+8]
8. What is a neural network? Explain the back propagation algorithms and perceptron. [2+4+4]
9. What is an expert system? Explain its advantages and disadvantages. [8]
10. What is a Natural Language Processing? Describe Natural Language Processing Steps and its application. [2+6]

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1. If the Turing Test is passed, does this show that computers exhibit intelligence? State your reasons. [7]
2. What is a Constraint satisfaction problem? Solve the following crypto-arithmetic problem with necessary steps:
LOGIC+LOGIC=PROLOG. [3+4]
3. Searching is an important part of AI, justify it. Explain any two types of blind search and compare them in terms of space and time complexity. [2+7]
4. What is a rule based reasoning? Explain Backward Chaining with suitable example. [7]
5. Using resolution solve the following statements:
All pompeian are Romans. All Romans were either loyal to Caesar or hated him.
Everyone is loyal to someone. People only try to assassinate rulers they not loyal to.
Marcus tried to assassinate Caesar. Marcus was pompeian. Find, did Marcus Caesar? [7]
6. What are Frames and Semantic Net? Convert the given sentences in semantic Net. [7]
 - i) A person is a mammal.
 - ii) Sakti Gauchan is a person.
 - iii) Person has nose.
 - iv) Sakti Gauchan is in Nepalese team.
 - v) Uniform color of sakti Gauchan is Red/Blue.
7. What is a machine learning? Explain genetic algorithm (GA) along with GA operators. List some of areas where GA can be applied. [2+6+2]
8. What do you understand by a perceptron? How can we design a neural network that acts as an OR gate. [2+7]
9. Differentiate between declarative knowledge and procedural knowledge. Describe expert system with its architecture and practical uses. [3+6]
10. Explain different steps involved in the natural language processing (NLP) with suitable block diagram and examples. [8]

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1. If the Turing Test is passed, does this show that computers exhibit intelligence? State your reasons. [7]
2. Solve the following puzzle by assigning numeral (0-9) in such a way that each letter is assigned unique digit which satisfy the following addition. [7]
ONE + ONE + TWO = FOUR
3. Explain the necessity of searching techniques in AI? Differentiate between Breadth first search and Depth first search with their performance criteria. [4+5]
4. Assume the following facts: [8]
 - i) Horses, cows, pigs are mammals
 - ii) An offspring of a horse is a horse
 - iii) Bluebeard is a horse
 - iv) Bluebeard is Charlie's parent
 - v) Offspring and parent are inverse relations
 - vi) Every mammal has a parent

Prove Charlie is a horse using resolution refutation.
5. What is causal net? How does Bayes Theorem calculate the probability in a causal net? Explain with example calculation. [7]
6. Convert given sentences into Semantic Network. [7]
 - i) The height of the adult male is 5.10
 - ii) Baseball player is an adult male.
 - iii) Adult male is a person.
 - iv) Batting average of Baseball players is 0.252
 - v) Pee-wee-Reese is a Fielder.
 - vi) Fielder is Baseball player.
 - vii) Team of pee-wee-Reese is Brooklyn Dodger.
7. "Learning is an essential characteristic for intelligent agents." List down justification on this statement. Write about the role learning with suitable example. [4+4]
8. What are applications of Expert System? Describe the Development stages of Expert System briefly. [2+6]
9. Define a NLU and a NLG. List down the different steps involved in the natural language processing (NLP) with suitable examples. [2+7]
10. Define Hebbian learning. Use Hebbian learning algorithm to Construct Hebbian Network which perform line AND Function. [3+7]

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1. Define Artificial Intelligence (AI). Describe the importance and practical application of AI. [6]
2. A farmer has a goat, a wolf and a cabbage on the west side of a river. He wants to get all of his animals and his cabbage across the river onto the east side. The farmer has a row boat but he only has enough room for himself and one other thing. The wolf will eat the goat if they are left together alone. The goat will eat the cabbage if they are left together alone. How can the farmer get everything on the east side? [8]
 - i) Formulate this puzzle as search
 - ii) Solve this problem-using search (any method)

Draw the search tree and show the final solution
3. Devise an example to show how A* algorithm uses path cost and heuristic cost to generate best solution. [8]
4. Consider the following axioms: [10]
 - i) Anyone whom Mary loves is a football star
 - ii) Any student who does not pass does not play
 - iii) John is a student
 - iv) Any student who does not study does not pass
 - v) Anyone who does not play is not a football star.

Prove that "If John does not study, then Mary does not love John" Resolution by Refutation.
5. A doctor is called to see a sick child. The doctor has prior information that 90% of sick children in that neighborhood have the flu, while the other 10% are sick with measles. Let F stand for an event of a child being sick with Flu and M stand for an event of a child being sick with measles. Assume for simplicity that there no other maladies in that neighborhood. A well-known (and common) symptom of measles is a rash and has probability of 0.95. [8]

However, very occasionally, children with flu also develop rash and has probability of 0.08.

Upon examining the child, the doctor finds a rash. What is the probability that the child has measles?
6. Explain Frames and Semantic Net with examples. List down their advantages and limitations. [4+4]
7. What is Fuzzy Logic and why is it important? Explain about Mamdani Fuzzy Inference Method with example. [3+7]
8. What do you understand by Perception? How can we design a neural network that acts as an XOR gate? [1+7]
9. Differentiate between declarative knowledge and procedural knowledge. Describe expert system with its architecture and practical uses. [3+5]
10. What s Natural Language processing? Explain the different steps in the natural language processing. [2+4]

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1. What is a rational agent? "System that think like humans" and "System that act like humans" are the part of artificial intelligence. Justify these statement with practical example. [1+6]
2. Define constraint satisfaction problem (CSP). Solve the following crypto-arithmetic problem, where different letters denote different integers and identical letters denote same integer. SWIM + WEAR = RELAX. [1+6]
3. Why Searching is important in problem solving? What do the drawbacks of greedy best-first search and how A* search algorithm solve it. Explain with an example. [2+7]
4. a) Assume the following facts: [8]
 - John likes all kinds of food.
 - Apples are food.
 - Chicken is food.
 - Anything anyone eats and isn't killed by is food.
 - Bill eats peanuts and is still alive.
 - Sue eats everything bill eats.

Prove that John likes peanuts using resolution.
- b) Differentiate between forward and backward chaining. [4]
5. What is Frame? How is it different from semantic net in knowledge representation? [7]
6. Define inductive Learning. Explain in detail about ID3 process with suitable example. [2+6]
7. What is self-organizing Map (SOM)? Explain all the steps involved in SOM with suitable example. [2+5]
8. Justify that the study of gene is one of important part in the AI. List down the steps involved in genetic algorithm with an example. [4+4]
9. How knowledge acquisition is performed in expert system? Explain one real expert system example with proper architecture. [2+5]
10. Write short notes on: (Any two) [2×4]
 - a) Machine Vision
 - b) Supervised Vs Unsupervised learning
 - c) Back Propagation Algorithm

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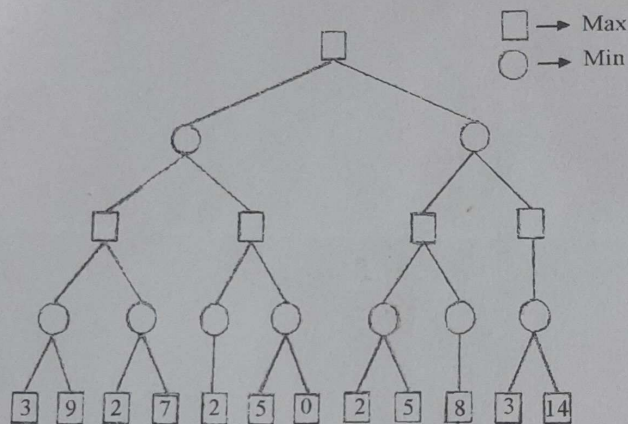
1. Define AI. What is the importance of Turing Test in AI? List applications of AI. [2+4+2]
2. What is problem space? Solve the following crypto arithmetic problem by showing all the steps. [2+6]
BASE + BALL = GAMES
3. Discuss the hill climbing search algorithm along with problem associated with it and discuss their solutions. Why simulated annealing is important? [6+2]
4. Given premises "Every American who sells weapons to hostile nations is a criminal. The country XYZ is enemy of America. All of its misallies in XYZ were sold by Donald, who is an American." Prove that Donald is a criminal by using FOPL based resolution refutation method. [8]
5. Why CNF is required? Explain all the steps used to convert a quantified statement with suitable example. [2+6]
6. Why semantic network and frames are important in AI? Provide examples of both with FOPL statements example. [2+6]
7. What is a genetic algorithm? Explain all steps in genetic algorithm with block diagram and operators. [8]
8. List the importance of expert system in real life. Draw block diagram of expert system architecture and explain all blocks. [2+6]
9. What is a McCulloch/Pitts neural network? Explain it with reference to AND gate. Justify that it cannot be applied to Exclusive OR gate. [8]
10. Justify that NLP is one of the important part of an AI. Explain the steps involved in the NLP. [8]

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1. Define Artificial Intelligence (AI)? Differentiate between Natural (Human) Intelligence and Artificial Intelligence. Can AI replace Humans? Justify. [1+3+4]
2. What is Constraint Satisfaction problem? Solve the following problem LOGIC + LOGIC = PROLOG. [3+5]
3. What is minmax search? Apply minmax search along with alpha-beta pruning for the following tree at max node at top. [2+6]



4. What is semantic network? Convert the following sentences into semantic network.
 - i) The avg height of an adult is 5'6".
 - ii) Cricket player is an adult male.
 - iii) Adult male is a person.
 - iv) Batting average of a cricket player is 37.59
 - v) Rohit Paudel is a Fielder.
 - vi) Fielder is a cricket player.
 - vii) Team of Rohit Paudel is Nepal.
5. Why CNF is necessary? "Everyone who loves all animals are loved by someone" represent this statement in FOPL and explain all the steps involved to convert it into CNF. [2+6]
6. What is belief network? How reasoning is done in this network? [2+6]
7. What is evolutionary computing? List down the steps in genetic algorithm with an example. [2+6]
8. What is perceptron? When do we need back propagation algorithm? Explain with an example. [2+6]
9. What is expert system? Create a scenario where we require expert system. List down the steps how you create this system. [2+6]
10. When do we need Natural Language Processing? Explain different steps of NLP. [3+5]

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Year / Part	IV / I	Time	3 hrs.

Subject: - Artificial Intelligence (CT 710)

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1. Define Artificial Intelligence and Intelligent agent. Differentiate between different types of intelligent agent with examples. [2+6]
2. What is constraint satisfaction problem? Solve the following cryptarithmic problem. [2+6]
TEN + TEN + FORTY = SIXTY
3. How searching is done in adversarial search? How A* search overcomes problem associated with Greedy Search? Explain with example. [2+6]
4. Consider the following axioms:
 - i) Ram likes all kinds of food.
 - ii) Apples and vegetables are food.
 - iii) Anything anyone eats and not killed is food.
 - iv) Anil eats peanuts and is still alive.
 - v) Ram eats everything that Anil eats.

Proof that "Ram likes peanuts" using resolution refutation. [8]
5. What do you understand by Bayesian Network? In a County, 51% of the adults are males and the other 49% are females. One adult is randomly selected for a survey involving credit card usage. [2+6]
 - i) Find the prior probability that the selected person is a male.
 - ii) It is later learned that the selected survey subject was smoking a cigar. Also, 9.5% of males smoke cigars, whereas 1.7% of females smoke cigars (based on data from the Substance Abuse and Mental Health Services Administration).

Use this additional information to find the probability that the selected subject is a male.
6. How a knowledge representation is evaluated? Explain semantic net and frame representation with examples. [3+5]
7. Explain all the steps in Fuzzy Learning with block diagram with an example. [8]
8. What are the problem associated with NLP? How syntactic and semantic analysis is done during natural language processing? Explain with example. [3+5]
9. Describe mathematical model of neural network. What does it means to train a neural network? Write an algorithm to its learning. [2+2+4]
10. Write short notes on: [2×4]
 - a) SOM
 - b) Expert Systems

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1. Define intelligent agent. Explain the factors that are required to pass the Turing Test. [2+6]
2. What is well-defined problem? Solve the following crypto arithmetic problem by showing all the steps. [2+6]
LOVE + LOVE = HATE
3. Compare the informed and uniformed search techniques. What strategies can be employed to address the issue of local maxima in Hill Climbing Search? [4+4]
4. Define skolemization with example. Consider the following axioms: [2+6]
 - a) Anyone passing his Engineering exams and winning the lottery is happy.
 - b) Anyone who studies or is lucky can pass all his exams.
 - c) Sneha did not study but she is lucky.
 - d) Anyone who is lucky wins the lottery.By using resolution refutation, prove that Sneha is happy?
5. What is prior probability and posterior probability? A doctor knows that the disease meningitis causes a patient to have a stiff neck, say, 80% of the time. The doctor also knows some unconditional facts: the prior probability that any patient has meningitis is 1/50000, and the prior probability that any patient has a stiff neck is 1%. Determine the probability of meningitis when a patient has a stiff neck. [2+6]
6. Compare semantic net and frame. How you convert semantic net into frame? Explain with example. [3+5]
7. What are different methods for learning? Explain the steps involved in a genetic algorithm with a flowchart and justify why mutation is important. [3+5]
8. What is perception? Construct a Hebbian Network that performs like AND Gate. [2+6]
9. What is expert system? Explain how inference is performed in expert system? [3+5]
10. Write short notes on: [2×4]
 - a) Fuzzy Logic
 - b) Natural Language Processing

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1. What is an intelligent agent? What type of model it works on? Explain PEAS for a Medical Diagnosis System. [2+2+3]
2. What do you understand by constraint satisfaction problem? Solve the following crypto-arithmetic problem. EAT+THAT = APPLE [2+6]
3. What are the problem associated with hill climbing search? Explain how simulated annealing overcome problem associated in hill climbing search. [3+5]
4. Assume the following facts:
 - a) Sita likes all kinds of food.
 - b) Momo and Burger are food.
 - c) Anything anyone eats and isn't killed by is food.
 - d) Bill eats mushroom and is still alive.
 - e) Sue eats everything Bill eats.Prove that "Sita likes mushroom" using resolution refutation method. [8]
5. What is Bayesian network? Describe with an appropriate example. How inference can be done using this network? [5+3]
6. What are the different knowledge representation models? Discuss about semantic nets with an example. [2+5]
7. Discuss the steps involved in genetic algorithm with example. [8]
8. What are the problem associated with NLP? How syntactic morphological and semantic analysis is done during natural language processing? Explain with example. [2+5]
9. What are major characteristics of expert system? Explain with example. Explain a general approaches for developing an expert system with a block diagram. [2+5]
10. Write short notes on: [3×4]
 - a) Fuzzy logic
 - b) Computer vision
 - c) A* Search.

TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
Examination Control Division
2079 Bhadra

Exam.	Regular		
Level	BE	Full Marks	80
Programme	BEI	Pass Marks	32
Year / Part	IV / I	Time	3 hrs.

Subject: - Artificial Intelligence (CT 710)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Assume suitable data if necessary.

1. What is Turing test? How is can be used to measure intelligence of machine? Explain. [2+5]
2. What are the steps of Problem solving? Assume you are given three empty jugs; a 3 – gallon, a 5-gallon and a 9-gallon, and a pump can fill water only in 3- gallon jug. How you get exactly 7 gallons of water in the 9- gallon jug? Formalize the problems; write down production rules and draw search tree to solve this water Jug problem. [2+1+2+3]
3. Different between Depth first search and Breadth first search with their performance criteria. [7]
4. Prove that "Charlie is a mammal" with given propositions. [8]
 - a) Cows, pigs and horses are mammal
 - b) The child of horse is a horse
 - c) Bluebeard is a horse
 - d) Bluebeard is charlie's father
 - e) Child and father are inverse relation
 - f) Every mammal has a father
5. Differentiate between forward chaining and backward chaining using suitable example. [7]
6. List the issues need to be consider in Knowledge Representation techniques. Convert given sentences into Semantic Network. [2+6]
 - a) Tom is a cat
 - b) Tom fights with rat
 - c) Tom is owned by Ram
 - d) Tom is black in color.
 - e) Cats like milk
 - f) Rats like cheese
 - g) The cat sat on the bed
 - h) A cat is a mammal
 - i) A Rat is an animal
 - j) All mammals are animals
 - k) Every mammal gives birth a baby
7. What do you mean by membership of an element in a fuzzy set? Explain the steps involved in Fuzzy Learning with suitable example. [2+6]
8. Define Artificial Neural Network. How non-linearity can be modelled through this ANN? Explain with example. [1+7]
9. List down the challenges faced by NLP. Why do you think machine vision is important in AI? Explain with example. [2+5]
10. Write short notes on: [3×4]
 - a) Hill climbing algorithm
 - b) SOM
 - c) Expert system